WASP-10b

R-1454

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_141028` · created 2026-08-01T09:55:46+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456958.8404 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1679 +/- 0.0087
Transit depth (Rp/Rs)^2	2.82 +/- 0.29 [%]
Orbital Inclination (inc)	89.81 +/- 0.55
Airmass coefficient 1 (a1)	1.262 +/- 0.015
Airmass coefficient 2 (a2)	-0.1028 +/- 0.0024
Scatter in the residuals of the lightcurve fit is	1.17 %
Best Comparison Star	None
Optimal Aperture	3.62
Optimal Annulus	11.22
Transit Duration (day)	0.0955 +/- 0.001

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1679 ± 0.0087 vs 0.1592 (deviation 1.0σ)
Duration fit vs published	0.0955 d vs 0.0946 d (deviation 0.92σ)
Mid-time O–C	-2.9 min
Depth SNR	19.3
Residual scatter	1.17 %

Light curve

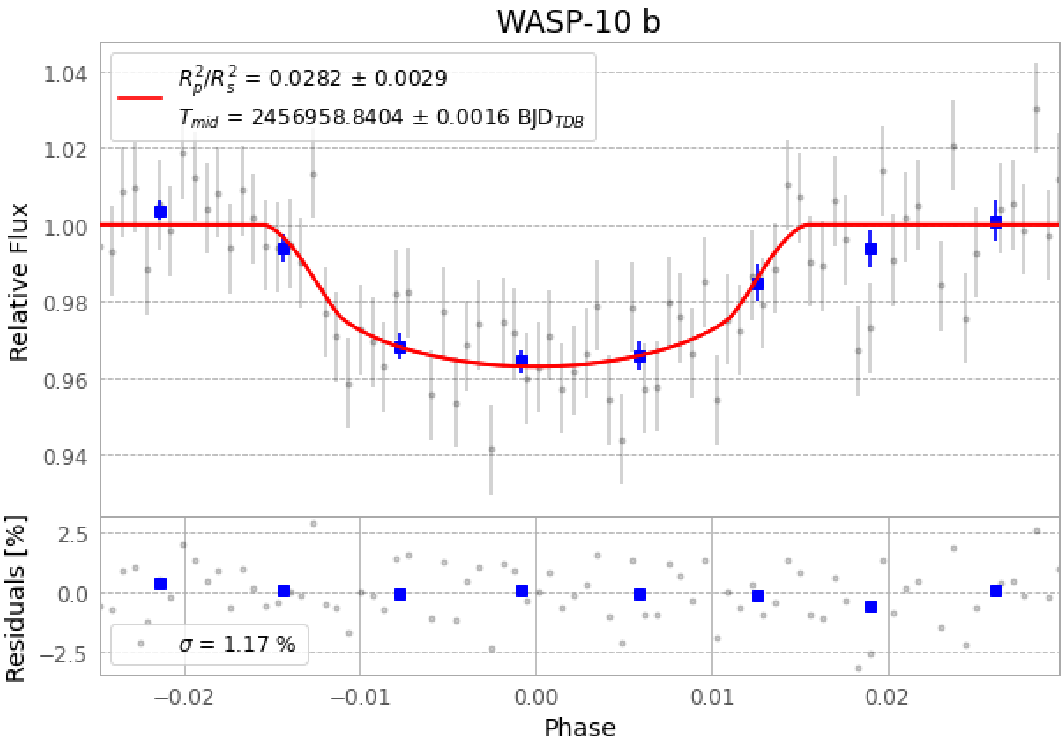

Figure 1 — FinalLightCurve_WASP-10 b_2014-10-27.png (SHA-256 2f42366a2a4e2d3c…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [206, 355], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 5cf461390196969f…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

82 FITS frames (54 MB), `WASP-10141028061213.FITS` → `WASP-10141028101512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-141028.log (ed3135e8e43b…), FinalLightCurve_WASP-10 b_2014-10-27.png (2f42366a2a4e…), FinalLightCurve_WASP-10 b_2014-10-27.csv (8045e1815254…)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 09:55Z.